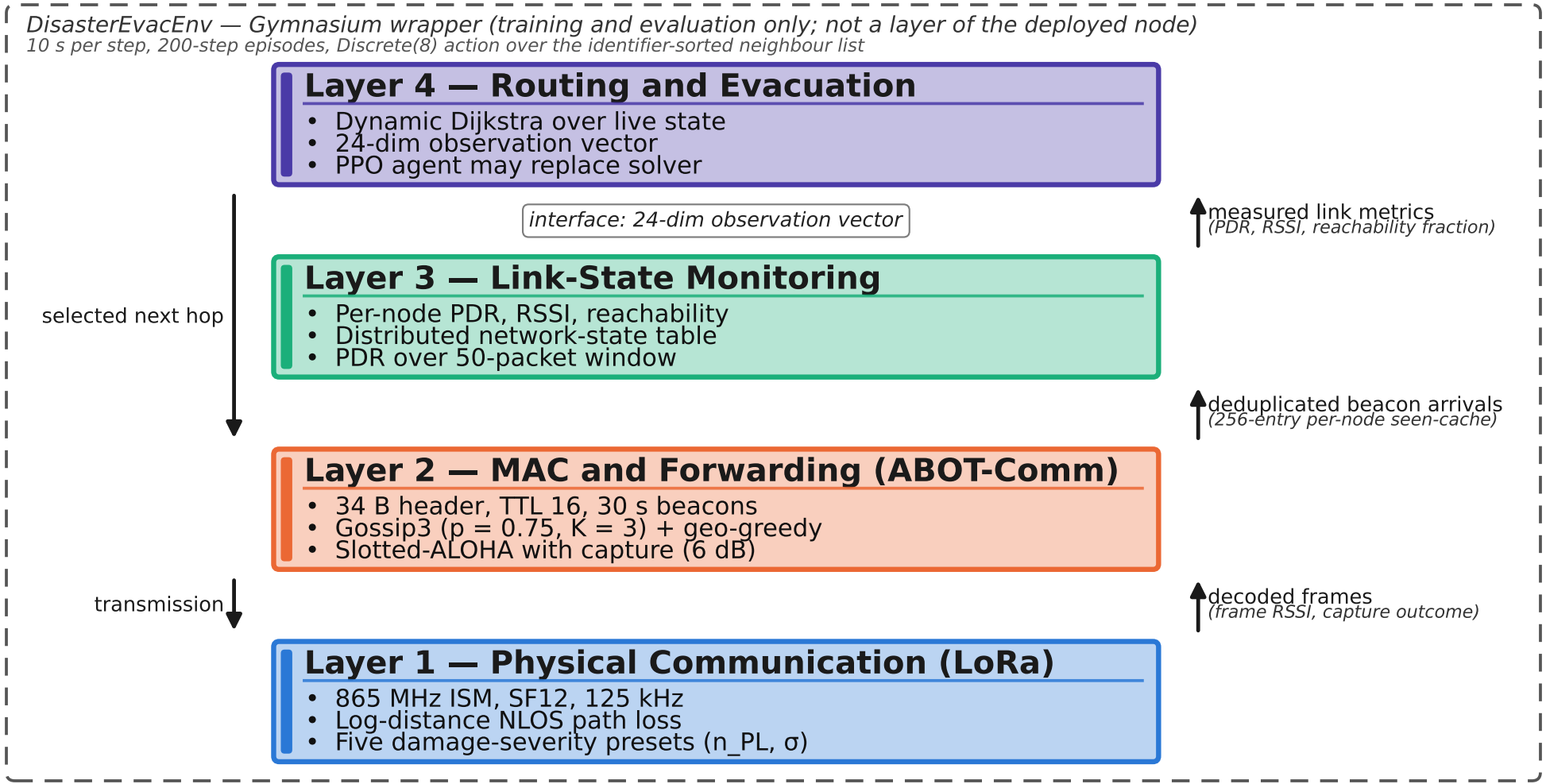
DisasterEvacEnv — Gymnasium wrapper (training and evaluation only; not a layer of the deployed node)
10 s per step, 200-step episodes, Discrete(8) action over the identifier-sorted neighbour list
Layer 4 — Routing and Evacuation
• Dynamic Dijkstra over live state
• 24-dim observation vector
• PPO agent may replace solver
interface: 24-dim observation vector
measured link metrics
(PDR, RSSI, reachability fraction)
selected next hop
Layer 3 — Link-State Monitoring
• Per-node PDR, RSSI, reachability
• Distributed network-state table
• PDR over 50-packet window
deduplicated beacon arrivals
(256-entry per-node seen-cache)
Layer 2 — MAC and Forwarding (ABOT-Comm)
• 34 B header, TTL 16, 30 s beacons
• Gossip3 (p = 0.75, K = 3) + geo-greedy
• Slotted-ALOHA with capture (6 dB)
transmission
decoded frames
(frame RSSI, capture outcome)
Layer 1 — Physical Communication (LoRa)
• 865 MHz ISM, SF12, 125 kHz
• Log-distance NLOS path loss
• Five damage-severity presets (n_PL, σ)